Study Guide for Population Growth and Demographics

This study guide provides examples for the topics that you might see on the first midterm from the lectures over population growth, and population demographics. This is NOT an exhaustive list of all content that might be covered on the exam. Use your notes from lecture, the lecture slides and video on Canvas, and to review all the topics that have been covered.

1. Determine whether each of the scenarios below describes a **density-dependent** or **density-independent** control on butterfly populations
    a. The females lay many eggs on each plant, such that the caterpillars eat nearly all the leaves and 80% of the caterpillars starve before becoming adults
    b. The nursery plant species on which the butterflies lay their eggs is entirely removed from the habitat
    c. A severe rainstorm drowns 80% of the butterflies
    d. When butterflies are not abundant, predators switch to eating moths, which reduces the mortality rate of butterflies
    e. A contagious disease sweeps through the population, killing 80% of the butterflies
    f. A late spring cold snap reduces temperatures to outside the butterflies' tolerance and 80% of them die in their cocoons

2. You are studying the population of squirrels on the UC Davis campus. You capture 53 squirrels across campus and "tag" them by putting little UC Davis sweaters on them, which are super adorable. The next week, you survey campus and count a total of 123 squirrels, 31 of which are wearing sweaters. Given these data, what is your estimate for the size of the UC Davis squirrel population?
    a. ~210 squirrels
    b. ~129 squirrels
    c. ~72 squirrels
    d. ~437 squirrels

3. Below is a table for the size of a population of yeast cells in a flask. What is *r* for this population across this time window? *Recall that the exponential growth equation is* $N_t = N_0e^{rt}$
    a. 0.747
    b. 0.617
    c. 0.768
    d. 0.434
    e. 0.482
    f. 0.609

Hour	Yeast Cells per mL of water
0	9
1	19
2	35.2
3	75.9
4	117.1
5	189.6

4. The life table below shows data for a fictional population of otters living in the UC Davis Arboretum waterway. Use the table and the provided equations to calculate or estimate the metrics below.

Age Class	Age in Years	Proportion surviving (l_x)	Age-specific fecundity (m_x)	$l_x m_x$	$x l_x m_x$
Newborn	0	1.0	0	0	0
Juvenile	1	0.97	0	0	0
Young adult	3	0.91	0.33	0.3	0.9
Adult	8	0.82	2.07	1.69	13.58
Senior Adult	14	0.43	0.98	0.42	5.89
TOTALS	---	---	---	2.41	20.37

$$R_0 = \Sigma l_x m_x \qquad G = \frac{\Sigma\, x\, l_x m_x}{\Sigma\, l_x m_x} \qquad r \approx \frac{(\ln R_0)}{G}$$

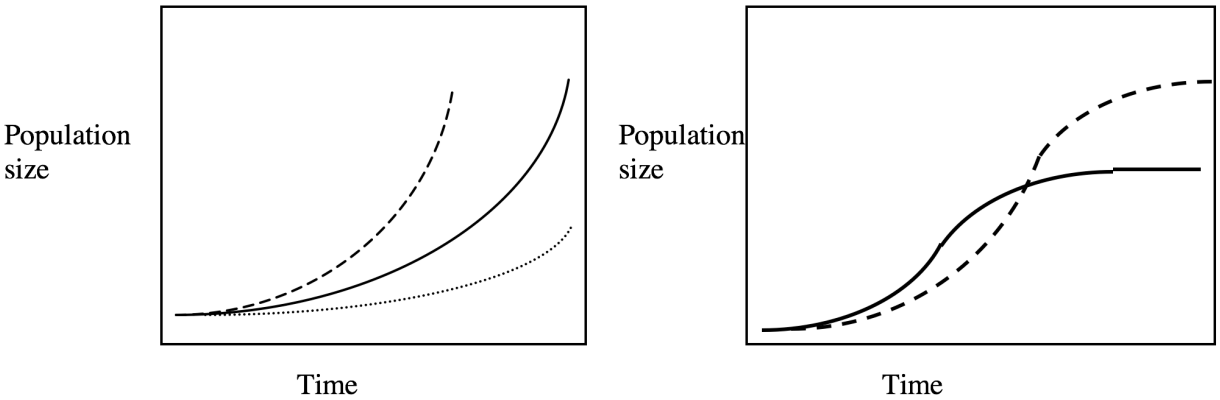
a. the average number of offspring an otter will have throughout its lifetime
b. the average age at reproduction
c. the per capita rate of growth

5. Below is a life table for a fictitious population of turtles in the Arboretum on the UCD campus. **Which of the statements below is best supported by the life table?**

Age class	Age in Years	Proportion surviving (l_x)	Age-specific fecundity (m_x)	$l_x m_x$	$x l_x m_x$
hatchlings	0	1.0	0	0	0
juvenile	1	0.25	0	0	0
Young adult	~3	0.20	3.0	0.6	1.8
Adult	~5	0.10	7.0	0.7	3.5
Old Adult	10+	0.10	0	0	0
TOTALS				1.3	5.3

a. This population has a Type I survivorship curve
b. The age class with the highest mortality also has the highest fecundity
c. On average, each individual in the population will have 5.3 offspring throughout its life
d. Fecundity increases throughout the entire life of the individual
e. None of the above are supported by this life table

6. Each graph below depicts population growth over time for different populations, shown by dotted, dashed, or solid curves. Answer the following questions using these two graphs



   a. Which of the graphs (left or right) describes exponential growth?
   b. On the left graph, which population has the highest *r*?
   c. On the right graph, which population has the highest K?
   d. On the right graph, which population has the highest *r*?

7. In a population of birds, we find that as the number of birds increases, the birth rate is decreasing and the death rate is increasing. This population is likely…
   a. Experiencing exponential growth
   b. Density independent
   c. At risk of extinction
   d. Approaching its carrying capacity
   e. Maintaining a stable population size

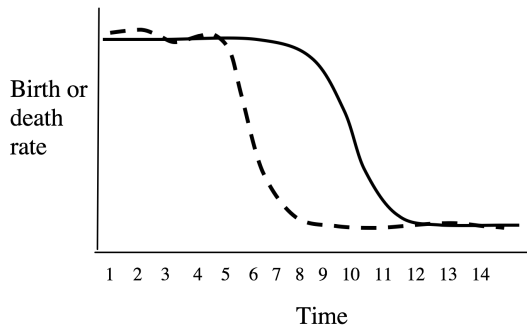
8. Below are data for a fictitious population of ducks at the Arboretum on the UCD campus. Use the data to answer the following questions. *(I've added in blank columns, which will help you in your calculations)*

Age class	Average Age in Years (x)	Proportion Surviving (l_x)	Age-specific Fecundity (m_x)	Lxmx	xlxmx
hatchlings	0	1.0	0		
juvenile	1	0.75	0		
Young adult	5	0.50	10		
Adult	7	0.25	12		
Old Adult	10+	0.01	0		

   a. Calculate the following metrics for this population:
      i. The net reproductive rate (R_0)
      ii. The generation time (*G*)
      iii. The per capita (or intrinsic) rate of increase (*r*)
   b. Is this population increasing or decreasing? How do you know?

9. Assume the populations below are experiencing exponential growth. The exponential growth equation is $N_t = N_0e^{rt}$
   a. For a population of 275 individuals with $r = 0.3$, what will the population size be after 3 years?
   b. For a population of 500 individuals with $r = 0.04$, what will the population size be after 3 years?
   c. For a population of 275 individuals with $r = 0.3$, how many years will it take for the population to double in size?
   d. How long will it take for a population with $r = 0.04$ to double in size?

10. Given the following patterns of birth (solid line) and death (dashed line) rates in a population over time, sketch a graph of population size through time (hint; you can start at any non-zero population size).



11. Assume the populations below are experiencing logistic growth. Round your answers to the nearest whole organism. The logistic growth equation is

$$N_t = \frac{K}{1+[(K-N_0)/N_0]e^{-rt}}$$

   a. For a population of 800 individuals with $r = 0.15$, living in an environment that can support 1000 individuals, what will the population size be after 1 year?
   b. For a population of 200 individuals with $r = 0.15$, living in an environment that can support 1000 individuals, what will the population size be after 1 year?
   c. Which of the populations from parts A and B above have a larger increase in population size in a single year? Explain why that population has a larger increase than the other population
   d. Is it possible biologically for a population to 'overshoot' the carrying capacity and have a larger population size than the environment can actually support? If so, what will happen to this population?

12. Explain the difference between r and R_o. Under which values of each is population growth positive, negative, and zero?

13. How can scientists estimate population sizes (N_o) of naturally occurring organisms? How can we estimate per capita growth rates (r)?

14. You are studying a population of feral cats which currently numbers 437 individuals. While the carrying capacity is approximately 500 cats for this region, the population is causing problems by hunting native birds and lizards and so your management goal is to get the population of cats down to 200 cats in the next 10 years.

    a. What *r* value would you need to achieve in order to hit this target?

    b. How many offspring, on average, can cats in this population have throughout their lifetime in order to stay below that *r* value? Assume generation time in cats is about 1 year.

15. You are studying a population of geckos in Hawai'i: you tag 15 geckos with unique paint markings and release them back into the habitat, then return a week later. On your second visit, you observe 40 geckos in the habitat, none of which bear a paint tag. Propose three explanations for why your mark-recapture studied failed.